(b) $y_n = n + \frac{100}{n}$ $\quad y_{n+1} = (n+1) + \frac{100}{n+1}$ $\quad y_{n+1} - y_n = 1 + \frac{100}{n+1} - \frac{100}{n}$

$$= \boxed{1 - \frac{100}{(n+1)n}}$$

$1 - \frac{100}{n^2+n} > 0 \Rightarrow (n^2+n) > 100$

$\Rightarrow \boxed{(n+10)(n-9) > 10}$

For L.H.S. to be positive, $n > 9$. $\boxed{n = 10, (n+10)(n-9) = 20 > 10}$

least no. for which the inequality holds

$\therefore$ Minimum value of sequence

$= y_{10} = 10 + \frac{100}{10} = 20$

3. Find the largest term of each sequence:

(a) $y_n = \frac{90n}{n^2+9}$ $\quad y_{n+1} = \frac{90(n+1)}{(n+1)^2+9} = \frac{90n+90}{n^2+2n+10}$ $\quad y_{n+1} - y_n = \frac{90n+90}{n^2+2n+10} - \frac{90n}{n^2+9}$

$= \frac{90n^2 + 810n + 90n^2 + 810 - 90n^2 - 180n^2 - 900n}{(n^2+2n+10)(n^2+9)}$

$= \frac{720n - 90n^2 + 810}{(n^2+2n+10)(n^2+9)} = \frac{810n - 90n - 90n^2 + 810}{(n^2+2n+10)(n^2+9)} = \frac{810(n+1) - 90n(n+1)}{(n^2+2n+10)(n^2+9)}$

$= \boxed{\frac{(n+1)90(9-n)}{(n^2+2n+10)(n^2+9)}}$

$n > 9$, the difference is negative

The largest term is $y_{10} = \frac{90 \times 10}{10^2+9} = \frac{900}{109}$

$= \frac{-90n - 90n^2 + 810}{(n^2+2n+10)(n^2+9)} = \frac{-90(n^2+n-9)}{(n^2+2n+10)(n^2+9)}$

$n^2+n-9 > 0$

$n(n+1) > 9$

$\boxed{n > 2}$

For $n > 2$, the difference is negative.

$\therefore$ Largest term $= y_3 = \frac{90 \times 3}{3^2+9} = \boxed{15}$

(b) $y_n = \frac{10^n}{n!}$ $\quad y_{n+1} = \frac{10^{n+1}}{(n+1)!}$ $\quad y_{n+1} - y_n = \frac{10^{n+1}}{(n+1)!} - \frac{10^n}{n!} = \frac{10^{n+1}(n!) - 10^n \cdot (n+1)!}{n!(n+1)!}$

$= \frac{10^n \cdot n!(10-n-1)}{n!(n+1)!} = \boxed{\frac{10^n(9-n)}{(n+1)!}} < 0$ for $n > 9$.

$\therefore$ Max value $= y_{10} = \frac{10^{10}}{10!} = y_9$